

Contactless Sleep Sensor

Communication Protocol

V1.18

Jun 5th 2025

HISTORY

Ref	Date	Change Reason
v1.0	6/30/2022	Creation
v1.1	8/13/2022	Add Device UUID
v1.2	11/17/2022	Add type and cmd description
v1.3	7/7/2023	Description for presence detection
v1.4	8/24/2023	Unified description
v1.5	8/31/2023	Unified description
v1.6	9/1/2023	Unified description
v1.7	9/13/2023	Add set in bed thresh
v1.9	11/21/2023	Add get in bed thresh
v1.12	3/25/2023	Add running time body energy
v1.13	4/2/92024	0x03e8 Add Body move energy and range
v1.14	11/19/2024	Add description
V1.15	4/21/2025	Add 0x03ed: In bed learning threshold
V1.16	5/26/2025	Add 0x0412: Reset radar parameters
V1.17	6/5/2025	Add WiFi Signal dBm (0xFF 0xFF); update overall description and supplement example codes

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1. Introduction

1.1. Purpose

This protocol aims to standardize the communication interface between the Contactless Sleep Sensor and the host, ensuring standardized and reliable data exchange. The primary objectives include:

- Defining a unified communication frame structure and data format
- Providing a complete command set and functional descriptions
- Supporting rapid integration for various application scenarios
- Ensuring system maintainability and extensibility

1.2. Updates and maintenance

We may update this document without previous announcement because of works related to product optimization. If you think your version is out of date, please consult factory to acquire the latest version.

2. Technical Specifications

This is a TCP protocol. This section provides detailed specifications for the communication interface and protocol format.

2.1. Frame Format

Field	Length	Type	Description	Example Value
Magic	1 byte	Hex	Fixed value: 0x13	13
Ver	1 byte	Hex	Fixed value: 0x01	01
Type	1 byte	Enum	Frame type: 0x00: response frame 0x01: request frame 0x02: one-way request frame	01
Cmd	1 byte	Enum	Command type: 0x00: Heartbeat 0x01: Request command 0x02: Response command	01
Req id	4 bytes	int	auto increment from 1 (for request-response matching)	00 00 00 23

Timeout	2 bytes	short	Timeout duration (0-32768)	00 0A
Content Len	4 bytes	int	Data field length, see section 3	00 00 00 14
Function Code	2 bytes	Hex	Command function identifier (see section 3)	00 01
Data Content	N bytes	Hex	Command-specific data (actual content, N=data length value, (see section 3)	00 00 00 00

Table 1 Basic Frame Format

2.2. Data Format Specifications

- Integer/Floating-point numbers: Big-endian (MSB first), e.g., 1.8 meters → 3F E6 66 66.
- **Status codes:**
0x00 0x00 0x00 0x00: Failed
0x00 0x00 0x00 0x01: Succeed
- **Firmware upgrade:** Segmented transmission (≤ 240 bytes per frame), with CRC16 verification

2.3. Communication Direction

Symbol	Description
↑	radar to host
↓	host to radar

Table 2 Communication Direction Symbols

3. Detailed Command Set Description

Function Code	Name	Remark
0x0001	Radar ID	Upload every second until a response is received and the connection is considered successful.
0x03E8	Vital Data	Radar reports data to the host.
0x03E9	Set Report Interval	ReportInterval.
0x03EA	Get Report Interval	
0x03EB	Set Working Distance	Working Distance
0x03EC	Get Working Distance	

0x03ED	Learn the bed-occupancy thresh	To learn the threshold of bed occupancy
0x03EE	Set No Breath Alert Timer	No Breath Alert Timer
0x03EF	Get No Breath Alert Timer	
0x03F0	No Breath Alert	
0x03F1	Set Low Breath Bpm Alert thresh	Low Breath Bpm Alert
0x03F2	Get Low Breath Bpm Alert thresh	
0x03F3	Low Breath Bpm Alert	
0x03F4	Set High Breath Bpm Alert thresh	High Breath Bpm Alert
0x03F5	Get High Breath Bpm Alert thresh	
0x03F6	High Breath Bpm Alert	
0x03F7	Set Low/Hight Breath Bpm Alert Timer	Time to trigger Low/Hight Breath Bpm Alert
0x03F8	Get Low/High Breath BPM Alert Timer	
0x03F9	Set No Heart Alert Timer	No Heart Alert Timer
0x03FA	Get No Heart Alert Timer	
0x03FB	No Heart Alert	
0x03FC	Set Low Heart Bpm Alert thresh	Low Heart Bpm Alert
0x03FD	Get Low Heart Bpm Alert thresh	
0x03FE	Low Heart Bpm Alert	
0x03FF	Set High Heart Bpm Alert thresh	High Heart Bpm Alert
0x0400	Get High Heart Bpm Alert thresh	
0x0401	High Heart Bpm Alert	
0x0402	Set Low/Hight Heart Bpm Alert Timer	Time to trigger Low/Hight Heart Bpm Alert
0x0403	Get Low/Hight Heart Bpm Alert Timer	
0x0404	Set Bed exit alert timer	Bed exit alert timer
0x0405	Get Bed exit alert timer	

0x0406	Bed exit alert	
0x040A	Turn on/off Turn-Over Alert	Turn on/off Turn-Over Alert
0x040B	Get the status of the Turn-Over Alert	
0x040C	Turn-Over alert	
0x040D	Turn on/off body movement	Turn on/off body movement
0x040E	Get the body movement counter	
0x040F	Report the energy for body movement	
0x0410	Get Radar ID	
0x0411	Reboot Radar	
0x0412	Reset radar parameters	
0x0418	Set In-bed detect thresh	In-bed detect thresh
0x0419	Get in-bed detect thresh	
0x0021	Start Upgrade	Upgrade steps
0x0022	Firmware file	
0x0023	Finish upgrade	
0xFFFF	WiFi Signal dBm	No response required

Table 3 List of the Command Set

3.1. Radar ID (0x0001)

● Command Description

The radar status will be automatically uploaded, refreshed every minute once the radar is successfully connected to the server.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x14	0x0001	↑

Data Content Structure:

Field	Type	Description	Remarks
radar type	u8	Radar type (breath\heart rate	

		0x01)	
hw version	U32	Hardware version	Big-endian
Device ID	int13	Unique identifier ID	

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 14 00 01 01 + [hw version] + [Unique Device ID]

● Request Example

Hardware version 03.00.00.24, Device ID: 13CECDA0000040C11D13155507)

13 01 01 01 00 00 00 01 27 10 00 00 00 14 00 01 01 03 00 00 18 13 CE CD A0 00 00 40 C1 1D 13 15 55 07

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0001	↓

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code	0x00 0x00 0x00 0x01:Acknowledgement

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 00 01 [Status code]

● Response Example (Success)

13 01 00 02 00 00 00 01 27 10 00 00 00 06 00 01 00 00 00 01

3.2. Vital Data (0x03E8)

● Command Description

The bio-vital data will be automatically uploaded to the server including breath BPM, heart rate BPM, body movement energy, etc.

● One-way Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x02	0x01	0x26	0x03E8	↑

Data Content Structure:

Field	Type	Description	Remarks
Breath BPM	Float	Respiratory rate (breaths per minute)	Example: 11.718
Breath curve	Float	Amplitude of respiratory waveform	Example: 0.152

Heart rate BPM	Float	Heart rate (beats per minute)	Example: 75
Heart rate curve	Float	Amplitude of heart rate waveform	Example: -0.428
Target distance	Float	Linear distance from the target to the radar	unit : m (Example: 1.5)
Signal strength	Float	Target signal strength (for bed occupancy detection)	For presence detection, if the signal strength is 0, there is nobody in bed.
Valid bit ID	Uint8	Data validity flag	valid_bit: Data validity ● 0: invalid breath/heart rate ● 1: valid breath ● 2: valid breath and heart rate
Body move energy	Float	Body movement energy (cumulative value)	Example: 22.472
Body move range	Float	Body movement amplitude (maximum value)	Example: 3.3

Complete structure of the request frame

```
13 01 02 01 [Req id] [Timeout] 00 00 00 26 03 E8 [Breath BPM] [Breath curve] [Heart rate BPM] [Heart rate curve] [Target distance] [Signal strength] [Valid bit ID] [Body move energy] [Body move range]
```

● Request Example

Breath BPM 11.718

Breath curve 0.152

Heart rate BPM 75

Heart rate curve -0.428

Target distance 1.5 m

Signal strength 35.912

Valid bit ID 2

Body Move Energy 22.472

Body Move Range 3.3

```
13 01 02 01 00 00 01 91 00 0A 00 00 00 26 03 E8 41 3B 80 00 3E 1C 0B C0 42 96 00 00 BE DB A4 F6 3F C0 00 00 42 0F A6 20 00 00 00 02 41 B3 C7 13 40 53 33 34
```

3.3. Set Report Interval (0x03E9)

● Command Description

The developer and set the time interval for the radar to report vital sign data using this command. The unit is 50 ms. For example, if set at 20, it means the data will be uploaded

every 1 second (20 * 50ms = 1000ms). The value must be between 1~60000 (corresponding to 50ms~3000s)..

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03E9	↓

Data Content Structure:

Field	Type	Description	Remarks
report interval	u8	Report interval (unit: 50ms)	Set value range : 1~60000 (frame repeat time) For example, if it is set at 20, the breath/heart rate is reported every 1 second.

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 E9 [report interval]

● Request Example (Set to 1 second: 20 × 50ms)

13 01 01 01 9B 00 00 00 10 27 00 00 00 06 03 E9 00 00 00 14

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03E9	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 E9 [Status code]

● Response Example (Success)

13 01 00 02 9B 00 00 00 00 00 00 00 06 03 E9 00 00 00 01

3.4. Get Report Interval (0x03EA)

● Command Description

Command to read the ReportInterval (unit: 50ms) . For example, if the value is 20, the breath/heart rate is reported every 1 second.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
------	-----	----------------	---------------	-----------

0x01	0x01	0x06	0x03EA	↓
------	------	------	--------	---

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 EA 00 00 00 00

● Request Example

13 01 01 01 9B 00 00 00 10 27 00 00 00 06 03 EA 00 00 00 00

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03EA	↑

Data Content Structure:

Field	Type	Description	Remarks
report interval	u8	Report interval (unit: 50ms)	Value range : 1~60000 (frame repeat time) For example, if it is set at 20, the breath/heart rate is reported every 1 second.

Complete structure of the response frame

12 01 00 02 [Req id] [Timeout] 00 00 00 06 03 EA [report interval]

● Response Example (Current interval is 1s: 20 ×50ms)

13 01 00 02 9B 00 00 00 00 00 00 00 06 03 EA 00 00 00 14

3.5. Set Working Distance (0x03EB)

● Command Description

The developer can use this command to set the Working Distance for the radar to search for the target. Once set, the radar will only detect targets within this range.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03EB	↓

Data Content Structure:

Field	Type	Description	Remarks
working_distance	float	Working distance. Unit: m. . For example, when set at 1.2, the radar will consider targets within a range of 0.9~1.2m as the chest of the user and use that signal to calculate the heart and breath data.	0: Automati search without limit. . >0: the max. Working Distance(unit:m). Valid: 0.9~3.0m. Default setting: 1.5. Unit: m. For example: 1.0m, 3F 80 00 00

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 EB [working_distance]

- **Request Example (1.2m)**

13 01 01 01 9B 00 00 00 10 27 00 00 00 06 03 EB 3F 99 99 9A

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03EB	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 EB [Status code]

- **Response Example (Success)**

13 01 00 02 9B 00 00 10 27 00 00 00 06 03 EB 00 00 00 01

3.6. Get Working Distance (0x03EC)

- **Command Description**

Command to read the Working Distance.

- **Request Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03EC	↓

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 EC 00 00 00 00

- **Request Example**

13 01 01 01 9B 00 00 00 00 00 00 06 03 EC 00 00 00 00

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03EC	↑

Data Content Structure:

Field	Type	Description	Remarks
working_distance	Float	To get the current working distance..	0: Automati search without limit. . >0: the max. Working

			Distance(unit:m).
--	--	--	-------------------

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 EC [working_distance]

● Response Example (1.5m)

13 01 00 02 9B 00 00 00 00 00 00 06 03 EC 3F C0 00 00

3.7. Learn the bed-occupancy thresh (0x03ED)

● Command Description

The developer can use this command to instruct the radar to learn the environment when the bed is empty (bed-occupancy thresh) so that the detection of bed occupancy can be more accurate. Once started, please make sure there is no one or other interference (such as moving objects) in or around the bed for at least one minute. The learning will take about one minute, once it's over, the radar will automaticall uploaded the thresh value it learned use it for bed-occupancy detection.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03ED	↓

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 ED 00 00 00 00

● Request Example

13 01 01 01 9B 00 00 00 00 00 00 06 03 ED 00 00 00 00

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03ED	↑

Data Content Structure:

Field	Type	Description	Remarks
In bed learning	Unit32	To learn the bed-occupancy thresh	

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 ED [In bed learning]

● Response Example (29.5dB)

13 01 00 02 9B 00 00 00 00 00 00 06 03 ED 41 EC 00 00

3.8. Set No Breath Alert Timer (0x03EE)

● Command Description

Command to set the alarm threshold for the duration of no-breath states. When the radar detects that the user has stopped breathing continuously for longer than the set threshold, it triggers an alarm.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03EE	↓

Data Content Structure:

Field	Type	Description	Remarks
threshold	int	NoBreathAlert threshold (unit: s)	Default setting: 0 0: turn off the function. 30~86400: the time threshold to trigger the NoBreathAlert.

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 EE [threshold]
```

● Request Example (120s)

```
13 01 01 01 9B 00 00 00 10 27 00 00 00 06 03 EE 00 00 00 78
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03EE	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01: Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 EE [Status code]
```

● Response Example (Success)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 03 EE 00 00 00 01
```

3.9. Get No Breath Alert Timer (0x03EF)

● Command Description

The developer can use this command to get the current setting to trigger the no breath alert. It is the time (in seconds) that the no breath alert will be triggered if no breath is detected within the entire time range.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03EF	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 EF 00 00 00 00
```

● Request Example

```
13 01 01 01 9B 00 00 00 00 00 00 00 06 03 EF 00 00 00 00
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03EF	↑

Data Content Structure:

Field	Type	Description	Remarks
threshold	int	NoBreathAlert threshold (unit: s)	0: turn off the function. 30~86400: the time threshold to trigger the NoBreathAlert.

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 EF [threshold]
```

● Response Example (120S)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 03 EF 00 00 00 78
```

3.10. No Breath Alert (0x03F0)

● Command Description

The radar can proactively reports no-breath alert events when no breath is constantly detected over a certain period of time (the threshold) This frame is automatic and the developer doesn't have to request anything. However, the following conditions must be met:

- 1) Breath BPM =0 in the **0x03 E8** data frame (zero breath data)
- 2) Duration of zero breath $\geq$ the time threshold set by **0x03EE**

● Report Frame Format

Type	Cmd	Content Length	Function Code	Direction
------	-----	----------------	---------------	-----------

0x01	0x01	0x06	0x03F0	↑
------	------	------	--------	---

Complete structure of the report frame:

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F0 00 00 00 00
```

● Report Example

```
13 01 01 01 00 00 00 00 00 00 00 00 06 03 F0 00 00 00 00
```

3.11. Set Low Breath Bpm Alert thresh (0x03F1)

● Command Description

To set the LowBreath(BPM)Alert threshold (unit: s).An alarm will be triggered when the detected respiratory rate keeps lower than the set value for longer than the set duration.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F1	↓

Data Content Structure:

Field	Type	Description	Remarks
threshold_bpm	int	Lowrespiratory rate alarm threshold (unit: BPM)	Default setting: 0 0: turn off the function. N: the LowBreath(BPM)Alert threshold. 0~1024 (BPM)

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F1 [threshold_bpm]
```

● Request Example (8 BPM)

```
13 01 01 01 9B 00 00 00 00 00 00 00 06 03 F1 00 00 00 08
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03F1	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 F1 [Status code]
```

● Response Example (Success)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 03 F1 00 00 00 01
```

3.12. Get Low Breath Bpm Alert thresh (0x03F2)

- **Command Description**

Command to read the LowBreath(BPM)Alert threshold (unit: s).

- **Request Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F2	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F2 00 00 00 00
```

- **Request Example**

```
13 01 01 01 9B 00 00 00 00 00 00 00 06 03 F2 00 00 00 00
```

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03F2	↑

Data Content Structure:

Field	Type	Description	Remarks
threshold_bpm	int	Current alarm threshold (unit: BPM)	Default setting: 0. 0: turn off the function. N: the LowBreath(BPM)Alert threshold.

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 F2 [threshold_bpm]
```

- **Response Example (8 BPM)**

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 03 F2 00 00 00 08
```

3.13. Low Breath Bpm Alert (0x03F3)

- **Command Description**

Once set, the radar will automatically send out the low breath alert when the detected value fulfills the set threshold. It must be used in conjunction with **0x03 F7** (alarm duration threshold). The following conditions must be met:

- 1) Breath BPM in **0x03E8** data frame < threshold set by **0x03F1**
- 2) Duration > time threshold set by **0x03 F7**

- **Report Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F3	↑

Complete structure of the report frame:

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F3 00 00 00 00

● Report Example

13 01 01 01 00 00 00 00 00 00 00 00 06 03 F3 00 00 00 00

3.14. Set High Breath Bpm Alert thresh (0x03F4)

● Command Description

To set the HighBreath(BPM)Alert threshold (unit: s).An alarm will be triggered when the detected respiratory rate keeps higher than the set value for longer than the set duration.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F4	↓

Data Content Structure:

Field	Type	Description	Remarks
threshold_bpm	int	High respiratory rate alarm threshold (unit: BPM)	Default setting: 0. 0: turn off the function. N: the HighBreath(BPM)Alert threshold. 0~1024 (BPM)

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F4 [threshold_bpm]

● Request Example (30 BPM)

13 01 01 01 9B 00 00 00 00 00 00 00 06 03 F4 00 00 00 1E

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03F4	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 F4 [Status code]

- **Response Example (Success)**

```
13 01 00 02 9B 00 00 00 00 00 00 00 00 06 03 F4 00 00 00 01
```

3.15. Get High Breath Bpm Alert thresh (0x03F5)

- **Command Description**

Command to read the HighBreath(BPM)Alert threshold (unit: s).

- **Request Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F5	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F5 00 00 00 00
```

- **Request Example**

```
13 01 01 01 9B 00 00 00 00 00 00 00 00 06 03 F5 00 00 00 00
```

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03F5	↑

Data Content Structure:

Field	Type	Description	Remarks
threshold_bpm	int	Current alarm threshold (unit: BPM)	Default setting: 0. 0: turn off the function. 1~1024: the respiratory BPM threshold.

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 F5 [threshold_bpm]
```

- **Response Example (30 BPM)**

```
13 01 00 02 9B 00 00 00 00 00 00 00 00 06 03 F5 00 00 00 1E
```

3.16. High Breath Bpm Alert (0x03F6)

- **Command Description**

Once set, the radar will automatically send out the high breath alert when the detected value fulfills the set threshold. It must be used in conjunction with **0x03F7** (alarm duration threshold). The following conditions must be met:

- 1) Breath BPM in **0x03E8** data frame > threshold set by **0x03F1**

2) Duration $\geq$ time threshold set by **0x03F7**

● Report Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F6	↑

Complete structure of the report frame:

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F6 **00 00 00 00**

● Report Example

13 01 01 01 00 00 00 00 00 00 00 06 03 F6 **00 00 00 00**

3.17. Set Low/High Breath Bpm Alert Timer (0x03F7)

● Command Description

To set the time interval to trigger the High/Low Breath(BPM) Alert (unit: s). Please note that the High/Low Breath(BPM) Alert will only be triggered when the detected respiratory rate is consistently higher/lower than the set value. It must be used in conjunction with 0x03F1 (low respiratory rate alarm) and 0x03F4 (high respiratory rate alarm).

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F7	↓

Data Content Structure:

Field	Type	Description	Remarks
time_threshold	Int	Duration threshold (unit: s)	30~86400: the time interval to trigger the High/Low Breath(BPM) Alert (default setting: 120).

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F7 [time_threshold]

● Request Example (120s)

13 01 01 01 9B 00 00 00 00 00 00 06 03 F7 **00 00 00 78**

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03F7	↑

Data Content Structure:

Field	Type	Description	Remarks
-------	------	-------------	---------

Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed
-------------	--------	-------------------------------	--

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 F7 [Status code]

● Response Example (Success)

13 01 00 02 9B 00 00 00 00 00 00 00 06 03 F7 00 00 00 01

3.18. Get Low/High Breath Bpm Alert Timer (0x03F8)

● Command Description

Command to read the time interval to trigger the High/Low Breath(BPM) Alert (unit: s).

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F8	↓

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F8 00 00 00 00

● Request Example

13 01 01 01 9B 00 00 00 00 00 00 00 06 03 F8 00 00 00 00

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03F8	↑

Data Content Structure:

Field	Type	Description	Remarks
time_threshold	Int	Current duration threshold (unit: s)	30~86400: the time interval to trigger the High/Low Breath(BPM) Alert (default setting: 120).

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 F8 [time_threshold]

● Response Example (120s)

13 01 00 02 9B 00 00 00 00 00 00 00 06 03 F8 00 00 00 78

3.19. Set No Heart Alert Timer (0x03F9)

● Command Description

To set the time threshold to trigger the the NoHeartAlert (unit:s). When no heart rate is detected over the entire set time range, the no heart alert will be triggered.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03F9	↓

Data Content Structure:

Field	Type	Description	Remarks
threshold	int	No-heart-rate alarm threshold (unit: s)	Default setting: 0. 0: turn off the function. 30~86400: the time threshold to trigger the NoHeartAlert.

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 F9 [threshold]
```

● Request Example (180s)

```
13 01 01 01 9B 00 00 00 10 27 00 00 00 06 03 F9 00 00 00 B4
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03F9	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 F9 [Status code]
```

● Response Example (Success)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 03 F9 00 00 00 01
```

3.20. Get No Heart Alert Timer (0x03FA)

● Command Description

Command to read the time threshold to trigger the NoHeartAlert.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03FA	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 FA 00 00 00 00
```

● Request Example

```
13 01 01 01 9B 00 00 00 00 00 00 00 00 06 03 FA 00 00 00 00
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03FA	↑

Data Content Structure:

Field	Type	Description	Remarks
threshold	int	Current alarm threshold (unit: BPM)	Default setting: 0. 0: turn off the function. 30~86400: the time threshold to trigger the NoHeartAlert.

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 FA [threshold]
```

● Response Example (180S)

```
13 01 00 02 9B 00 00 00 00 00 00 00 00 06 03 FA 00 00 00 B4
```

3.21. No Heart Alert (0x03FB)

● Command Description

Once set, the radar will automatically send out the no heart alert when the detected value fulfills the set threshold. The following conditions must be met:

- 1) Heart rate BPM=0 in the **0x03E8** data frame (zero heart rate data)
- 2) No-heart-rate duration > time threshold set by **0x03F9**

● Report Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03FB	↑

Complete structure of the report frame:

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 FB 00 00 00 00
```

● Report Example

```
13 01 01 01 00 00 00 00 00 00 00 00 00 06 03 FB 00 00 00 00
```

3.22. Set Low Heart Bpm Alert thresh (0x03FC)

● Command Description

To set the LowHeartAlert (BPM) threshold (unit: s). An alarm is triggered when the detected heart rate remains lower than the set value for longer than the set time range.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03FC	↓

Data Content Structure:

Field	Type	Description	Remarks
threshold_bpm	int	Low heart rate alarm threshold (unit: BPM)	Default setting: 0. 0: turn off the function. N: the LowHeartAlert (BPM) Alert threshold. 1~1024 (BPM)

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 FC [threshold_bpm]
```

● Request Example (50 BPM)

```
13 01 01 01 9B 00 00 00 00 00 00 00 06 03 FC 00 00 00 32
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03FC	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 FC [Status code]
```

● Response Example (Success)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 03 FC 00 00 00 01
```

3.23. Get Low Heart Bpm Alert thresh (0x03FD)

● Command Description

Command to read the LowHeartAlert (BPM) threshold (unit: s).

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03FD	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 FD 00 00 00 00
```

● Request Example

```
13 01 01 01 9B 00 00 00 00 00 00 00 00 06 03 FD 00 00 00 00
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03FD	↑

Data Content Structure:

Field	Type	Description	Remarks
threshold_bpm	int	Current alarm threshold (unit: BPM)	0: turn off the function. N: the LowHeartAlert (BPM) threshold.

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 FD [threshold_bpm]
```

● Response Example (48 BPM)

```
13 01 00 02 9B 00 00 00 00 00 00 00 00 06 03 FD 00 00 00 30
```

3.24. Low Heart Bpm Alert (0x03FE)

● Command Description

Once set, the radar will automatically send out the low heart alert when the detected value fulfills the set threshold. The following conditions must be simultaneously met:

- 1) Heart rate BPM in the **0x03E8** data frame < threshold BPM set by **0x03FC**
- 2) Duration > time threshold set by **0x0402**

● Report Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x03FE	↑

Complete structure of the report frame:

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 FE 00 00 00 00
```

● Report Example

```
13 01 01 01 00 00 00 00 00 00 00 00 00 06 03 FE 00 00 00 00
```

3.25. Set High Heart Bpm Alert thresh (0x03FF)

● Command Description

To set the HighHeart(BPM)Alert threshold (unit: s). Once set, the radar will automatically send out the high heart alert when the detected value fulfills the set threshold. It must be used

with 0x0402 to set the duration threshold. Please note that this command cannot be triggered together with low heart rate alarm (0x03FE).

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x00	0x01	0x06	0x03FF	↓

Data Content Structure:

Field	Type	Description	Remarks
threshold_bpm	int	High heart rate alarm threshold (unit: BPM)	Default setting: 0. 0: turn off the function. N: the HighHeart(BPM)Alert threshold. 1~1024 (BPM)

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 03 FF [threshold_bpm]
```

● Request Example (120 BPM)

```
13 01 01 01 9B 00 00 00 00 00 00 00 06 03 FF 00 00 00 78
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x03FF	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 03 FF [Status code]
```

● Response Example (Success)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 03 FF 00 00 00 01
```

3.26. Get High Heart Bpm Alert thresh (0x0400)

● Command Description

Command to read the HighHeart(BPM)Alert threshold (unit: s).

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0400	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 00 00 00 00 00
```

- **Request Example**

```
13 01 01 01 9B 00 00 00 00 00 00 00 00 06 04 00 00 00 00 00
```

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0400	↑

Data Content Structure:

Field	Type	Description	Remarks
threshold_bpm	int	Current alarm threshold (unit: BPM)	0: turn off the function. N: the HighHeart(BPM)Alert threshold. 1~1024

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 00 [threshold_bpm]
```

- **Response Example (120 BPM)**

```
13 01 00 02 9B 00 00 00 00 00 00 00 00 06 04 00 00 00 00 78
```

3.27. High Heart Bpm Alert (0x0401)

- **Command Description**

Once set, the radar will automatically send out the high heart alert when the detected value fulfills the set threshold. The following conditions must be simultaneously met:

- 1) Heart rate BPM in the **0x03E8** data frame > threshold BPM set by **0x03FF**
- 2) Duration > time threshold set by **0x0402**

- **Report Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0401	↑

Complete structure of the report frame:

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 01 00 00 00 00
```

- **Report Example**

```
13 01 01 01 00 00 00 00 00 00 00 00 00 06 04 01 00 00 00 00
```

3.28. Set Low/High Heart Bpm Alert Timer (0x0402)

- **Command Description**

To set the time interval to trigger the High/Low Heart (BPM) Alert (unit: s). Please note that the High/Low Heart (BPM) Alert will only be triggered when the detected heart rate is consistently higher/lower than the set value. It must be used in conjunction with **0x03FC** (Set low heart BPM alert thresh) and **0x03FF** (Set high heart BPM alert thresh).

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0402	↓

Data Content Structure:

Field	Type	Description	Remarks
time_threshold	Int	Duration threshold (unit: seconds)	30~86400: the time interval to trigger the High/Low Heart (BPM) Alert

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 02 [time_threshold]
```

● Request Example (90s)

```
13 01 01 01 9B 00 00 00 00 00 00 00 06 04 02 00 00 00 5A
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0402	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 02 [Status code]
```

● Response Example (Success)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 04 02 00 00 00 01
```

3.29. Get Low/High Heart Bpm Alert Timer (0x0403)

● Command Description

Command to read the time interval to trigger the High/LowHeartAlert Timer (BPM) Alert (unit: s).

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0403	↓

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 03 00 00 00 00

● Request Example

13 01 01 01 9B 00 00 00 00 00 00 06 04 03 00 00 00 00

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0403	↑

Data Content Structure:

Field	Type	Description	Remarks
time_threshold	Int	Current duration threshold (unit: s)	30~86400 : the time interval to trigger the High/Low Heart (BPM) Alert.

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 03 [time_threshold]

● Response Example (120s)

13 01 00 02 9B 00 00 00 00 00 00 06 04 03 00 00 00 78

3.30. Set Bed exit alert timer (0x0404)

● Command Description

The developer than use this command to set the time threshold to trigger the Bed Exit Alert (unit:s). The bed exit alert will be automatically triggered once the user is away from the bed and the bed is empty for longer than the set threshold.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x00	0x01	0x06	0x0404	↓

Data Content Structure:

Field	Type	Description	Remarks
threshold_time	int	Bed exit alarm threshold (unit: s)	Default setting: 0. 0: turn off the function. N: the time threshold to trigger the Bed Exit Alert. 30~86400 (second)

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 04 [threshold_time]

- **Request Example (65 s)**

13 01 01 01 9B 00 00 00 00 00 00 00 06 04 04 00 00 00 41

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0404	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 04 [Status code]

- **Response Example (Success)**

13 01 00 02 9B 00 00 00 00 00 00 00 06 04 04 00 00 00 01

3.31. Get Bed exit alert timer (0x0405)

- **Command Description**

The developer can use this command to read the time threshold to trigger the Bed Exit Alert (unit:s).

- **Request Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0405	↓

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 05 00 00 00 00

- **Request Example**

13 01 01 01 9B 00 00 00 00 00 00 00 06 04 05 00 00 00 00

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0405	↑

Data Content Structure:

Field	Type	Description	Remarks
threshold_time	int	Bed exit alarm threshold (unit: s)	0: turn off the function. 30~86400: the time threshold to

			trigger the Bed Exit Alert
--	--	--	----------------------------

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 05 [threshold_time]

● Response Example (120 s)

13 01 00 02 9B 00 00 00 00 00 00 06 04 05 00 00 00 78

3.32. Bed exit alert (0x0406)

● Command Description

Once set, the radar will automatically sent the bed exit alert when the thresholds are met. The conditions to trigger the alert:

- 1) The radar detects that the bed is empty (Breath BPM and Heart rate BPM both being 0 in the 0x03E8 data frame indicates bed exit).
- 2) The bed has been empty for longer than the set time threshold (0x0404).

● Report Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0406	↑

Complete structure of the report frame:

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 06 00 00 00 00

● Report Example

13 01 01 01 00 00 00 00 00 00 00 06 04 06 00 00 00 00

3.33. Turn on/off Turn-Over Alert (0x040A)

● Command Description

The developer can use the turn-over alert to decide if the user has moved in bed, usually for ulcer prevention. Use this command to turn on / off the Turn-Over Alert. When turned on, no-turn-over alarms can be triggered via 0x040C.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x040A	↓

Data Content Structure:

Field	Type	Description	Remarks
enable_flag	Unit8	Turn-over detection function switch	Default setting: 0. 0: turn off the function.

			1: algorithm based on users turning over in bed.
--	--	--	--

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 0A [enable_flag]

● Request Example (Turn on)

13 01 01 01 9B 00 00 00 00 00 00 00 06 04 0A 00 00 00 01

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x040A	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 0A [Status code]

● Response Example (Success)

13 01 00 02 9B 00 00 00 00 00 00 00 06 04 0A 00 00 00 01

3.34. Get the status of the Turn-Over Alert (0x040B)

● Command Description

Command to read the status of the Turn-Over Alert .

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x040B	↓

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 0B 00 00 00 00

● Request Example

13 01 01 01 9B 00 00 00 00 00 00 00 06 04 0B 00 00 00 00

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x040B	↑

Data Content Structure:

Field	Type	Description	Remarks
-------	------	-------------	---------

enable_flag	Unit8	Current status of turn-over detection	0 : turn off the function. 1 : algorithm based on users turning over in bed.
-------------	-------	---------------------------------------	---

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 0B [enable_flag]
```

● Response Example (Turned on)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 04 0B 00 00 00 01
```

3.35. Turn-Over alert (0x040C)

● Command Description

Once set, the radar will automatically send the turn-over alert when the thresholds are met.

Please turn on the function via **0x040A**. The alert will be triggered if:

- 1) Body movement detection (**0x040D**) is turned on
- 2) User turning-over motion is detected

● Report Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x040C	↑

Complete structure of the report frame:

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 0C 00 00 00 00
```

● Report Example

```
13 01 01 01 00 00 00 00 00 00 00 00 06 04 0C 00 00 00 00
```

3.36. Turn on/off body movement (0x040D)

● Command Description

The developer can use this command to turn ON/OFF detection for body movement. It will affect the turn-over alert and the detection of body movement energy.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x040D	↓

Data Content Structure:

Field	Type	Description	Remarks
enable_flag	Unit8	Body movement detection switch	Default setting: 1. 0: OFF

			1: ON
--	--	--	-------

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 0D [enable_flag]

● Request Example (On)

13 01 01 01 9B 00 00 00 00 00 00 00 06 04 0D 00 00 00 01

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x040D	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 0D [Status code]

● Response Example (Success)

13 01 00 02 9B 00 00 00 00 00 00 00 06 04 0D 00 00 00 01

3.37. Get the body movement counter (0x040E)

● Command Description

Command to read the body movement counter.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x040E	↓

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 0E 00 00 00 00

● Request Example

13 01 01 01 9B 00 00 00 00 00 00 00 06 04 0E 00 00 00 00

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x040E	↑

Data Content Structure:

Field	Type	Description	Remarks
-------	------	-------------	---------

enable_flag	Unit8	Current status of body movement detection	0: OFF 1: ON
-------------	-------	---	-----------------

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 0E [enable_flag]

● Response Example (On)

13 01 00 02 9B 00 00 00 00 00 00 00 06 04 0E 00 00 00 01

3.38. Report the energy for body movement (0x040F)

● Command Description

The data reported value is the accumulated energy value for every body movement. Only when body movements detected will the radar automatically upload the body movement energy, which is the accumulated value for each event. If no movement is detected, the radar will send nothing. To enable this function, the following condition must be met:

- 1) Body movement detection (0x040D) is turned on

● Report Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x02	0x01	0x06	0x040F	↑

Data Content Structure:

Field	Type	Description	Remarks
Body move energy	Float * 4	the accumulated energy for every body movement.	Higher values indicate stronger user movement amplitude

Complete structure of the report frame:

13 01 02 01 [Req id] [Timeout] 00 00 00 06 04 0F [Body move energy]

● Report Example (95)

13 01 02 01 00 00 00 00 00 00 00 06 04 0F 42 BE 00 00

3.39. Get Radar ID (0x0410)

● Command Description

Please use this command to get the radar device's unique identifier ID for device identification and management.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0410	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 10 00 00 00 00
```

- **Request Example**

```
13 01 01 01 00 00 00 01 00 00 00 00 00 06 04 10 00 00 00 00
```

- **Response Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x0F	0x0410	↑

Data Content Structure:

Field	Type	Description	Remarks
radar_id	Unit8	Radar unique ID	

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 0F 04 10 [radar_id]
```

- **Response Example (13CECDA0000040C11D13155507)**

```
13 01 00 02 00 00 00 01 00 00 00 00 00 0F 04 10 13 CE CD A0 00 00 40 C1 1D 13 15 55 07
```

3.40. Reboot Radar (0x0411)

- **Command Description**

Please use this command to soft reset the radar. The hardware will not be reset. After a soft reset, it takes about 5 seconds to reconnect to the server.

- **Request Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0411	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 11 00 00 00 00
```

- **Request Example**

```
13 01 01 01 00 00 00 01 00 00 00 00 00 06 04 11 00 00 00 00
```

- **Response Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0411	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed

			0x00 0x00 0x00 0x01:Succeed
--	--	--	-----------------------------

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 11 [Status code]

● Response Example (Success)

13 01 00 02 00 00 00 01 00 00 00 00 00 06 04 11 00 00 00 01

3.41. Reset radar parameters (0x0412)

● Command Description

Please use this command to reset the radar parameters. Please be careful when enable this command as the radar will be reset to factory default settings.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0412	↓

Complete structure of the request frame

13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 12 00 00 00 00

● Request Example

13 01 01 01 00 00 00 01 00 00 00 00 00 06 04 12 00 00 00 00

● Response Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0412	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 12 [Status code]

● Response Example (Success)

13 01 00 02 00 00 00 01 00 00 00 00 00 06 04 12 00 00 00 01

3.42. Set In-bed detect thresh (0x0418)

● Command Description

Please use this command to manually set the in bed thresh, the threshold used for bed occupancy detection. The lower the value, the more sensitive the detection, and the higher the false alert rate.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0418	↓

Data Content Structure:

Field	Type	Description	Remarks
threshold	Float	Signal strength threshold	10-70

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 18 [threshold]
```

● Request Example (30.0)

```
13 01 01 01 9B 00 00 00 00 00 00 00 06 04 18 41 F0 00 00
```

● Response Frame Format (Req id matches request sequence number)

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0418	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Status code (success/failure)	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 18 [Status code]
```

● Response Example (Success)

```
13 01 00 02 9B 00 00 00 00 00 00 00 06 04 18 00 00 00 01
```

3.43. Get in-bed detect thresh (0x0419)

● Command Description

Please use this command to get the in-bed thresh.

● Request Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0419	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 04 19 00 00 00 00
```

● Request Example

```
13 01 01 01 9B 00 00 00 00 00 00 00 06 04 19 00 00 00 00
```

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0419	↑

Data Content Structure:

Field	Type	Description	Remarks
threshold	Float	Current signal strength threshold	

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 04 19 [threshold_bpm]
```

- **Response Example (30.0)**

```
13 01 00 02 9B 00 00 00 00 00 00 06 04 19 41 F0 00 00
```

3.44. Upgrade steps

Upgrade steps: 1. Send the upgrade package and the radar responds; 2. Send the firmware in segments; 3. Complete the upgrade.

3.44.1. Start Upgrade (0x0021)

- **Command Description**

Please use this command for the radar to get ready for the update. It carries the total bytes of the firmware to be updated.

- **Request Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x0021	↓

Data Content Structure:

Field	Type	Description	Remarks
Firmware size	U32	Size of the firmware	524292 bytes for example

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 00 21 [Firmware size]
```

- **Request Example (524292)**

```
13 01 01 01 00 00 00 01 00 00 00 00 06 00 21 00 08 00 04
```

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x0021	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Radar responds: Successful/Failed.	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 00 21 [Status code]
```

- Response Example (Success)**

```
13 01 00 02 00 00 00 01 00 00 00 00 00 06 00 21 00 00 00 01
```

3.44.2. Firmware file (0x0022)

- Command Description**

To send the firmware in segments. Each frame requires a CRC16 verification (only for the data portion).

- Request Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	Variable (N)	0x0022	↓

Data Content Structure:

Field	Type	Description	Remarks
Firmware	UInt8	Firmware content (Max allowance for the content: 240bytes.)	240 bytes or remaining bytes in the last frame
crc	uint16	CRC16 verified for every frame.	Only the firmware content, without the header, length and the func tag.

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] [Content Len] 00 22 [Firmware size]
```

- Request Example (240bytes)**

```
13 01 01 01 00 00 00 01 00 00 00 00 04 01 00 22 01 20 83 90 01 46 B3 F1 58 00 07 D0 17 38 09 D0 40
1E 03 D0 08 38 18 BF 0A 22 04 D1 10 22 02 E0 FF FF 0F 00 08 22 70 2B 81 98 00 F0 B3 80 43 F6 60 76
B8 46 30 40 20 38 B9 46 00 F0 8D 80 20 38 75 D0 C0 38 5C D0 B0 F5 80 70 42 D0 B0 F5 C0 60 31 D0
B0 F5 00 60 20 D0 B0 F5 80 50 0F D0 B3 F1 58 00 72 D0 0C 38 00 F0 8C 80 40 1F 00 F0 89 80 80 1F
6A D0 80 1F 68 D0 C0 1E 66 D0 94 E0 B3 F1 58 00 62 D0 0C 38 7C D0 40 1F 7A D0 80 1F 5C D0 80
1F 5A D0 C0 1E 58 D0 86 E0 B3 F1 58 00 54 D0 0C 38 6E D0 40 1F 6C D0 80 1F 4E D0 80 1F 4C D0
C0 1E 4A D0 78 E0 B3 F1 58 00 2F D0 0C 38 2D D0 40 1F 2B D0 80 1F 29 D0 80 1F 27 D0 C0 1E 25
D0 6A E0 B3 F1 58 00 0A D0 0C 38 52 D0 40 1F 50 D0 80 1F 04 D0 80 1F 18 BF B0 F1 03 00 5C D1 E4
1C 24 F0 03 04 24 1D 14 F8 04 8C 4F F0 00 09 52 E0 B3 F1 58 00 0A D0 0C 38 08 D0 40 1F 06 D0 80
1F 04 D0 80 1F 18 BF B0 F1 03 00 45 D1 E4 1D 24 F0 07 04 08 34 A4 F1 08 00 90 E8 00 03 3B E0 B3
F1 58 00 0A D0 0C 38 24 D0 40 1F 22 D0 80 1F 04 D0 80 1F 18 BF B0 F1 03 00 2E D1 E4 1C 24 F0 03
04 24 1D 54 F8 04 8C D0 E7 B3 F1 58 00 0A D0 0C 38 0F D0 40 1F 0D D0 80 1F 04 D0 80 1F 18 BF B0
F1 03 00 19 D1 E4 1C 24 F0 03 04 24 1D 34 F8 04 8C BB E7 E4 1C 24 F0 03 04 24 1D 54 F8 04 5C 4F
EA E5 79 07 E0 E4 1C 24 F0 03 04 24 1D 4F F0 00 09 54 F8 04 5C A8 46 90 94 C8 17 4C 46 45 46 54
EA 00 06 45 EA 01 00 08 BF 00 28 81 98 02 D1 01 09 C0 F0 43 81 64 2B 18 BF 69 2B 08 D1 00 2C 06
DA 6D 42 4F F0 00 00 60 EB 04 04 01 26 8E 96 85 A8 00 90 21 46 28 46 1B F0 13 FC 83 99 40 1C 9D
```

```
F8 10 32 81 42 85 99 29 DA 78 2B 03 D0 58 2B 18 BF 70 2B 11 D1 81 98 01 09 0F D3 70 2B 08 BF 78
23 85 98 30 21 03 70 85 98 40 1E 85 90 01 70 85 98 9D F8 10 32 40 1E 85 90 81 98 6F 2B 40 F0 0D 81
01 09 85 99 C0 F0 09 81 4A 78 30 2A 00 F0 05 81 30 20 08 70 85 98 40 1E 85 90 BF E0 30 22 0A 70 85
99 49 1E 85 91 C8 E7 E4 1D 24 F0 07 04 08 34 90 94 18 46 A0 F1 61 01 1A 29 14 ED 02 0B 38 BF 20
38 98 42 39 46 08 BF 01 21 59 EC 10 8B 47 F6 F0 74 47 F6 F0 70 04 EA 19 44 A0 42 4D 46 14 D1 91
9E 28 46 00 00 4F F0 00 0E 4E
```

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	Variable (N)	0x0022	↑

Data Content Structure:

Field	Type	Description	Remarks
Status code	Unit32	Responds from the radar of whether the frame is successful or not.	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed

Complete structure of the response frame

```
13 01 00 02 [Req id] [Timeout] 00 00 00 06 00 22 [Status code]
```

- **Response Example (Success)**

```
13 01 00 02 00 00 00 01 00 00 00 00 00 06 00 22 00 00 00 01
```

3.44.3. Finish upgrade (0x0023)

- **Command Description**

This command means the update has been completed, and it will trigger the radar to verify and adopt to the new firmware. If interrupted, the firmware can be resent to restart the upgrade.

- **Request Frame Format**

Type	Cmd	Content Length	Function Code	Direction
0x01	0x01	0x06	0x00 0x23	↓

Complete structure of the request frame

```
13 01 01 01 [Req id] [Timeout] 00 00 00 06 00 23 00 00 00 00
```

- **Request Example**

```
13 01 01 01 00 00 00 01 00 00 00 00 00 06 00 23 00 00 00 00
```

- **Response Frame Format (Req id matches request sequence number)**

Type	Cmd	Content Length	Function Code	Direction
0x00	0x02	0x06	0x00 0x23	↑

Data Content Structure:

Field	Type	Description	Remarks
-------	------	-------------	---------

Status code	Unit32	Radar responds: Successful/Failed.	0x00 0x00 0x00 0x00: Failed 0x00 0x00 0x00 0x01:Succeed
-------------	--------	---------------------------------------	--

Complete structure of the response frame

13 01 00 02 [Req id] [Timeout] 00 00 00 06 00 23 [Status code]

● Response Example (Success)

13 01 00 02 00 00 00 01 00 00 00 00 00 06 00 23 00 00 00 01

3.45. WiFi Signal dBm (0xFF 0xFF)

● Command Description

Please use this command to get the WiFi signal strength (dBm). It is an unidirectional notification, and no response is required.

● Report Frame Format

Type	Cmd	Content Length	Function Code	Direction
0x02	0x01	0x06	0xFF 0xFF	↑

Complete structure of the report frame:

13 01 02 01 [Req id] [Timeout] 00 00 00 06 FF FF 00 00 00 00

● Report Frame Example (-58dBm)

13 01 02 01 9B 00 00 00 10 27 00 00 00 06 FF FF FF FF FF C6